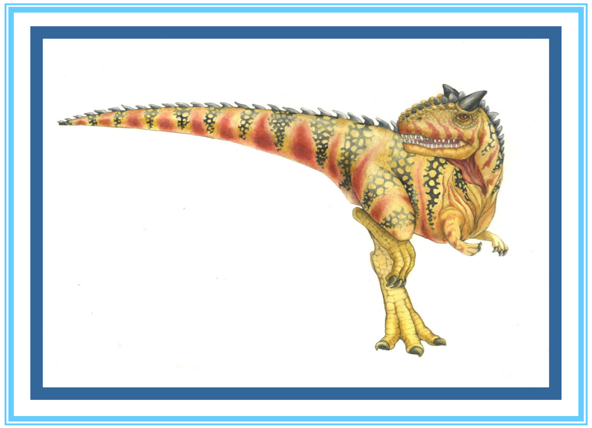
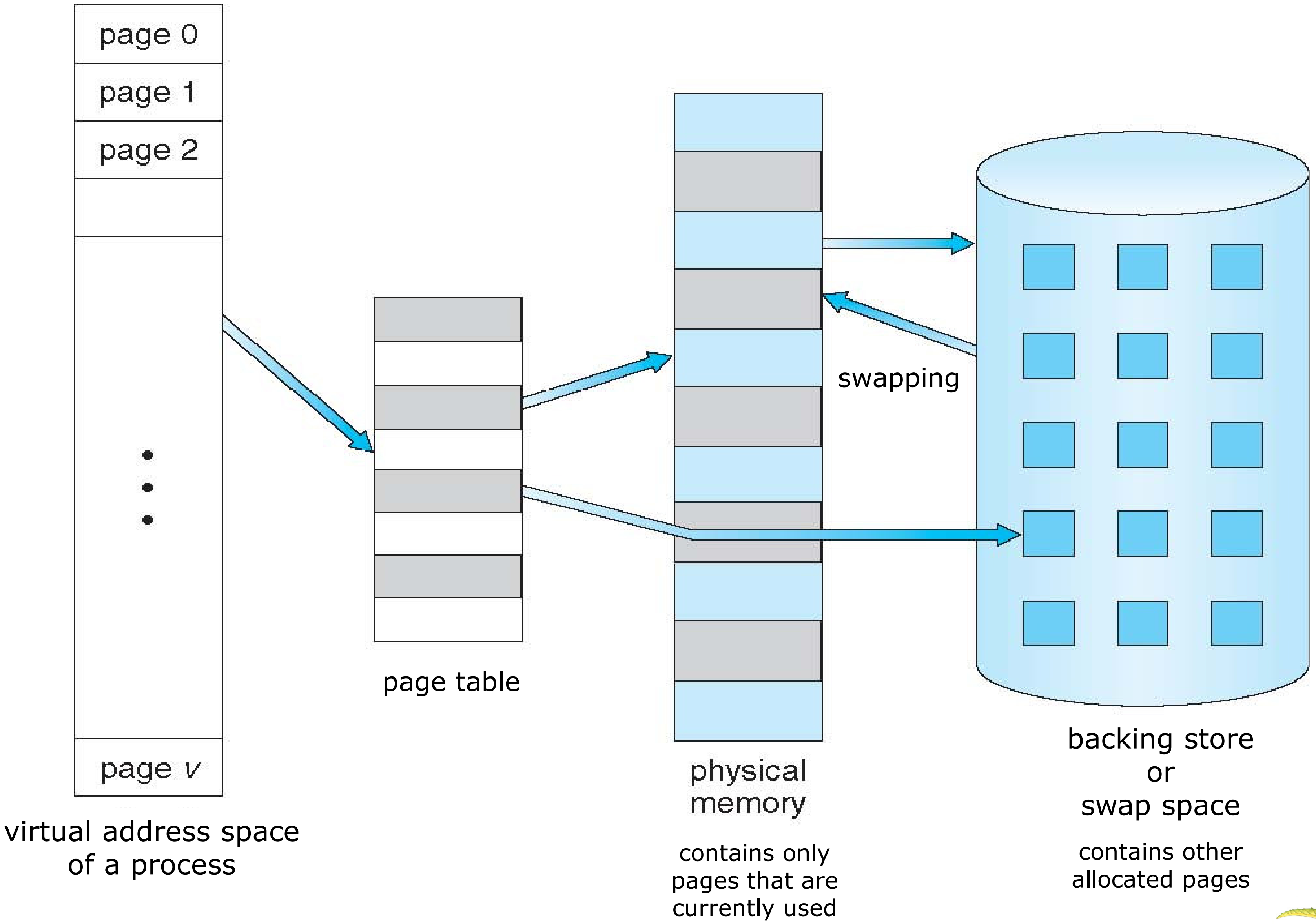
Chapter 9: Virtual-Memory Management



Virtual Memory

- Goal
  - Provide memory space for all processes that can be much larger than physical memory.
- How
  - Only some part of program code and data need to be in physical memory at a time.
    - principle of locality
  - The rest of program code and data can be stored in secondary storage (hard disk)
    - “swap partition” or “swap file”
    - extend memory hierarchy
  - Implement “Demand Paging” mechanism in OS
    - similar to “cache miss”
- Other benefits
  - Address spaces can be shared by several processes
    - shared program code, shared libraries, shared memory (IPC)
  - fast process creation (allocate memory only when needed)
  - memory-mapped file, memory-mapped I/O

Virtual Memory



Demand Paging

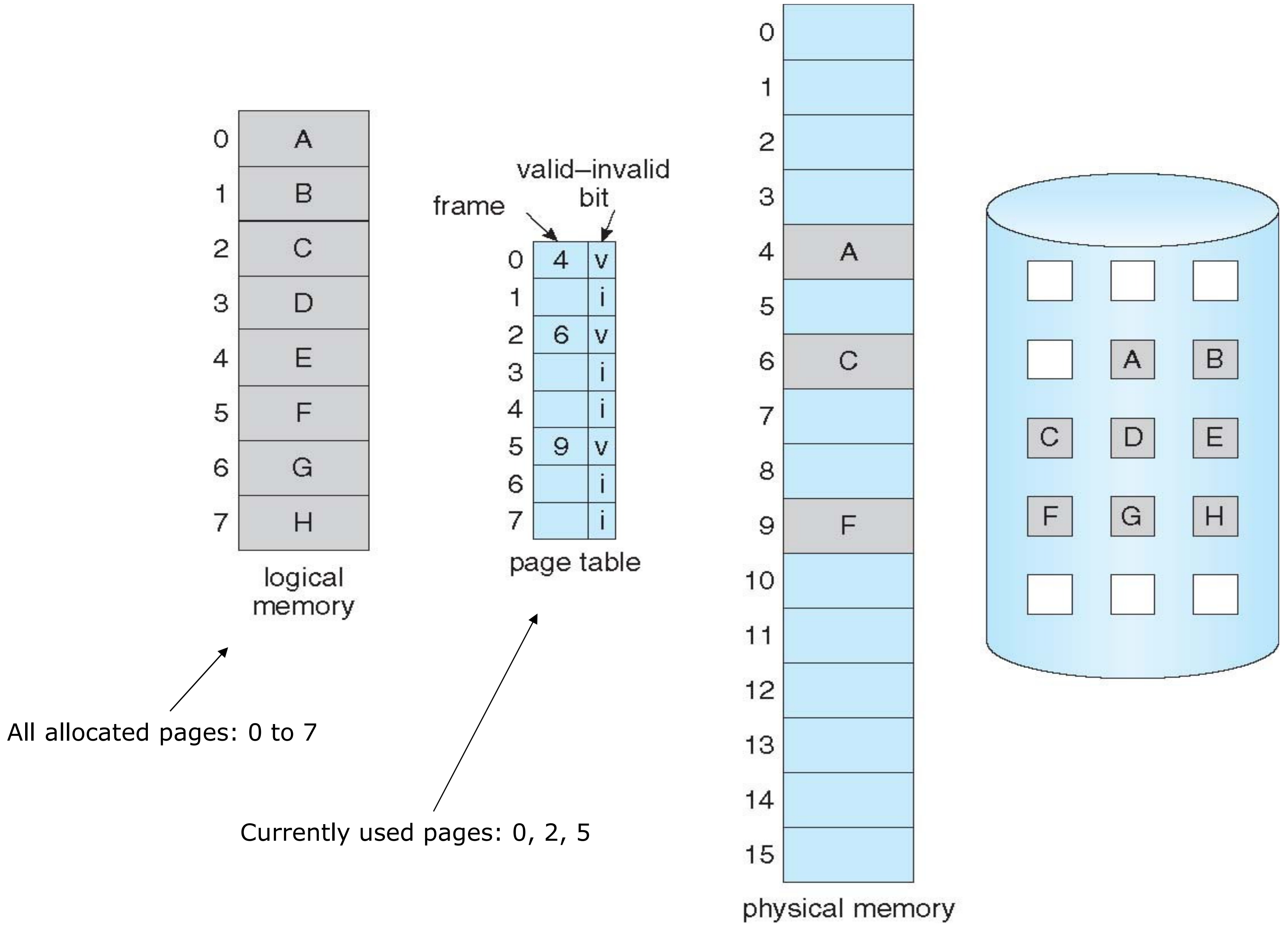
Bring a page into memory only when it is needed

- Less I/O needed
- Less memory needed
- Faster response
- More processes

CPU instruction references to anywhere in a page ⇒ the page needs to be in physical memory

- invalid reference ⇒ abort
- not-in-memory ⇒ bring to memory
- no-free-frame ⇒ swap

Page table that supports demand paging



Page Fault

Page fault is a mechanism to implement Demand Paging

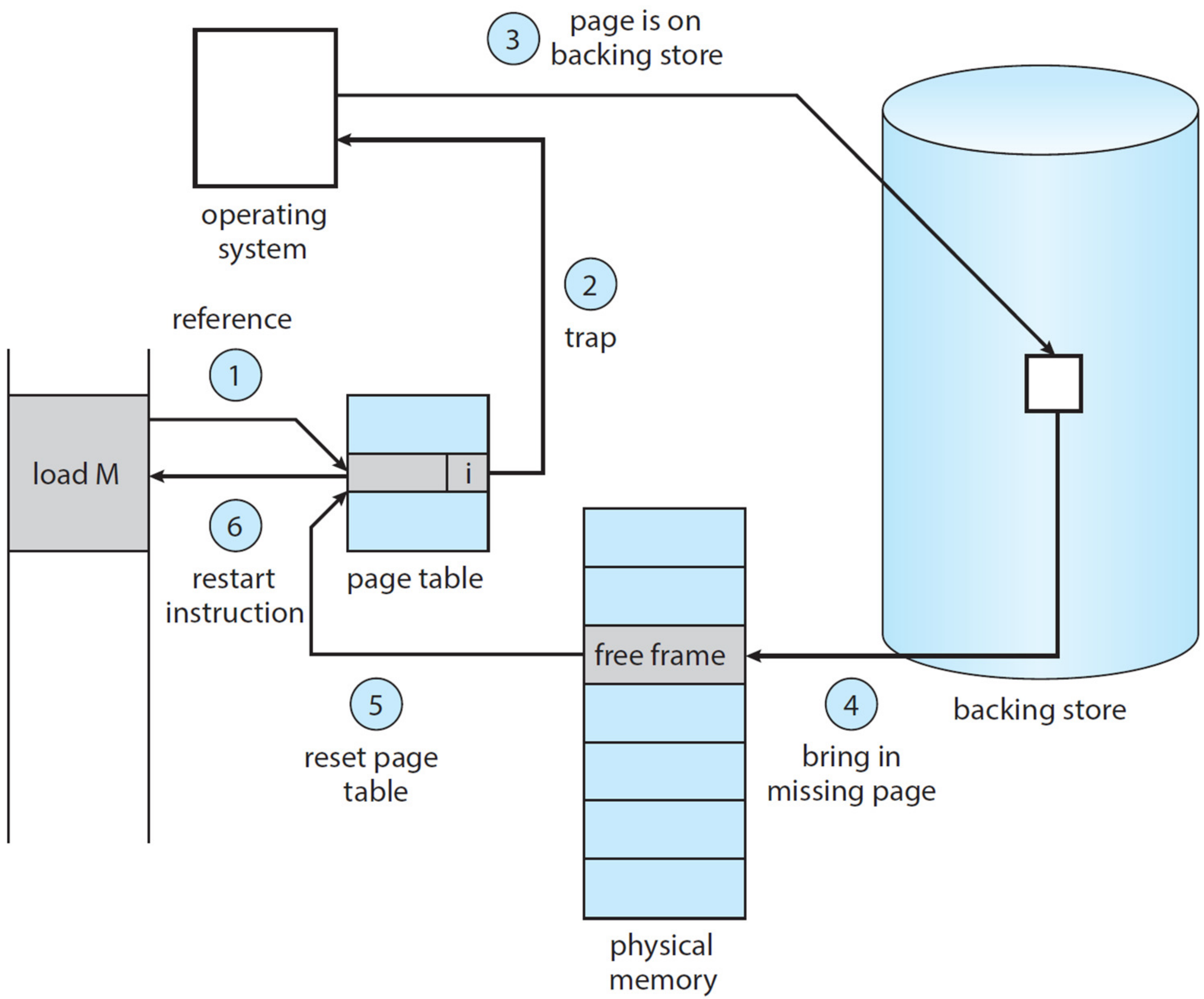
- Similar to cache miss

A reference to an invalid page (e.g. first reference to that page) will cause an interrupt to operating system:

- page fault interrupt (generated by MMU)
- page fault handler (code in OS kernel)

1. Operating system looks at another table to decide:
   - Invalid reference → abort
   - Just not in memory → continue
2. Get empty frame
3. Swap page into frame
4. Update tables
5. Set validation bit = **v**
6. Restart the instruction that caused the page fault

Steps in Handling a Page Fault

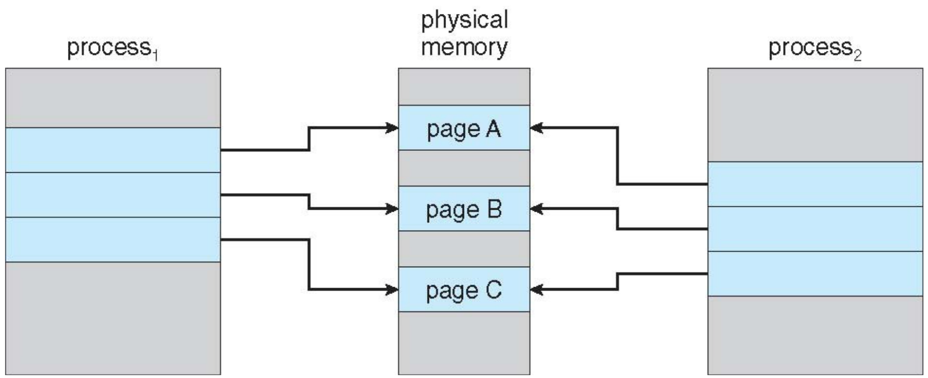


Copy-on-Write

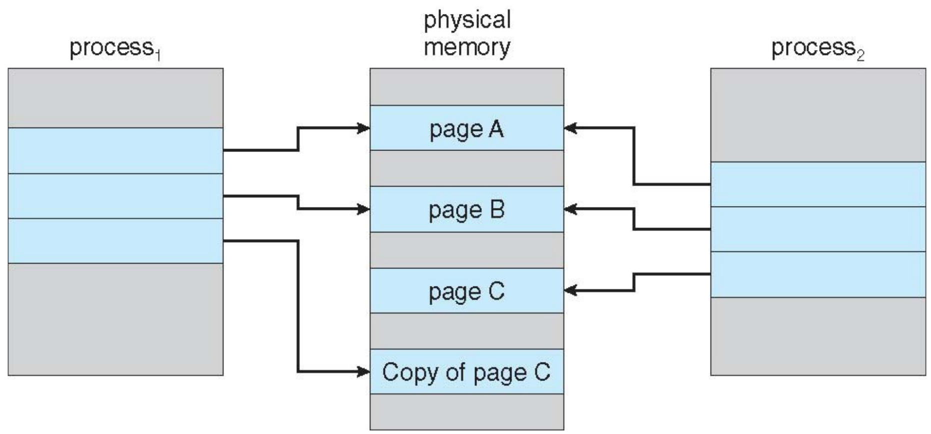
Copy-on-Write (COW) allows both parent and child processes to initially *share* the same pages in memory

- If either process modifies a shared page, only then is the page copied

COW allows more efficient process creation as only modified pages are copied

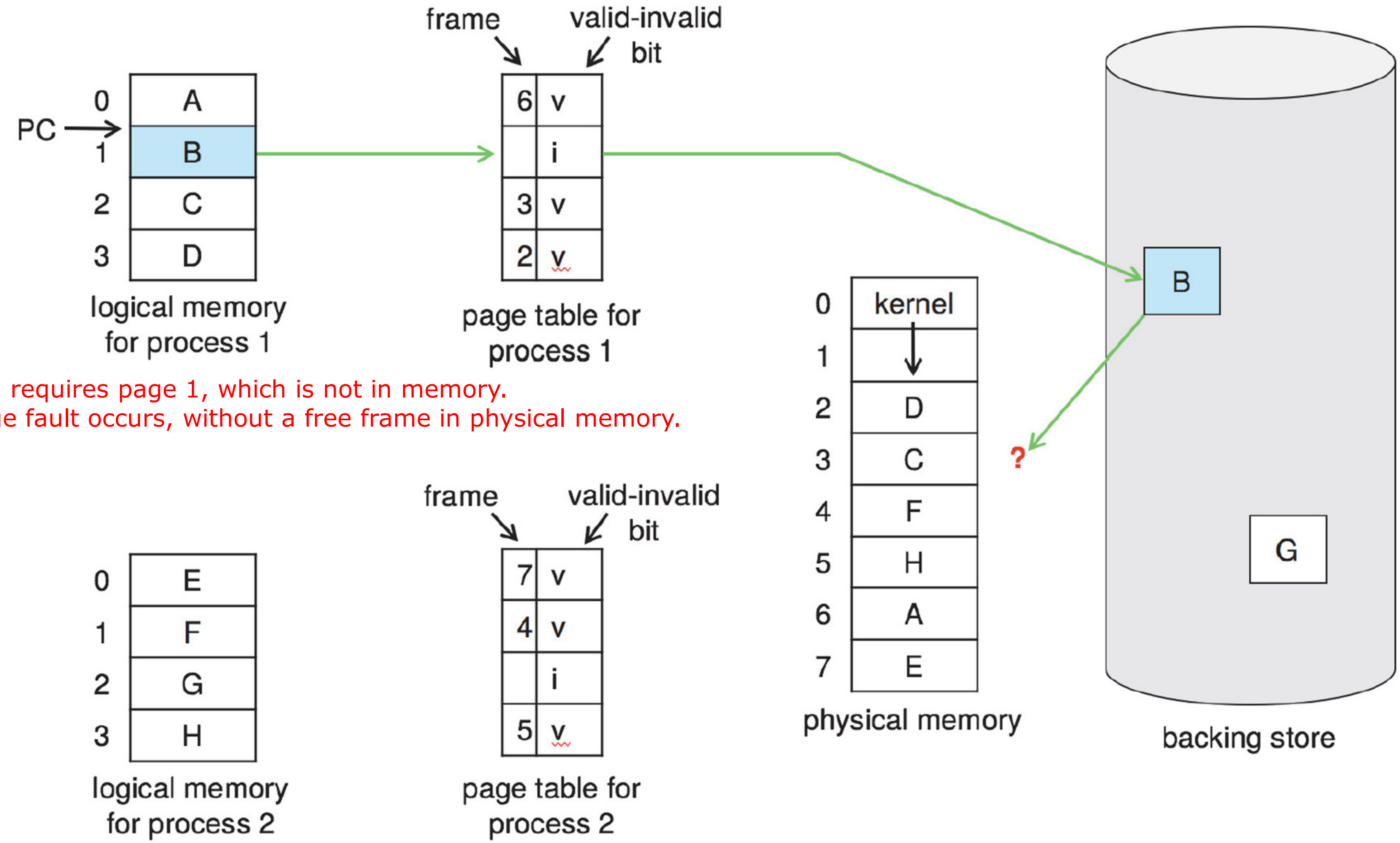


Before Process 1 Modifies Page C



After Process 1 Modifies Page C

Need For Page Replacement



Process 1 requires page 1, which is not in memory.
So, a page fault occurs, without a free frame in physical memory.

Basic Page Replacement

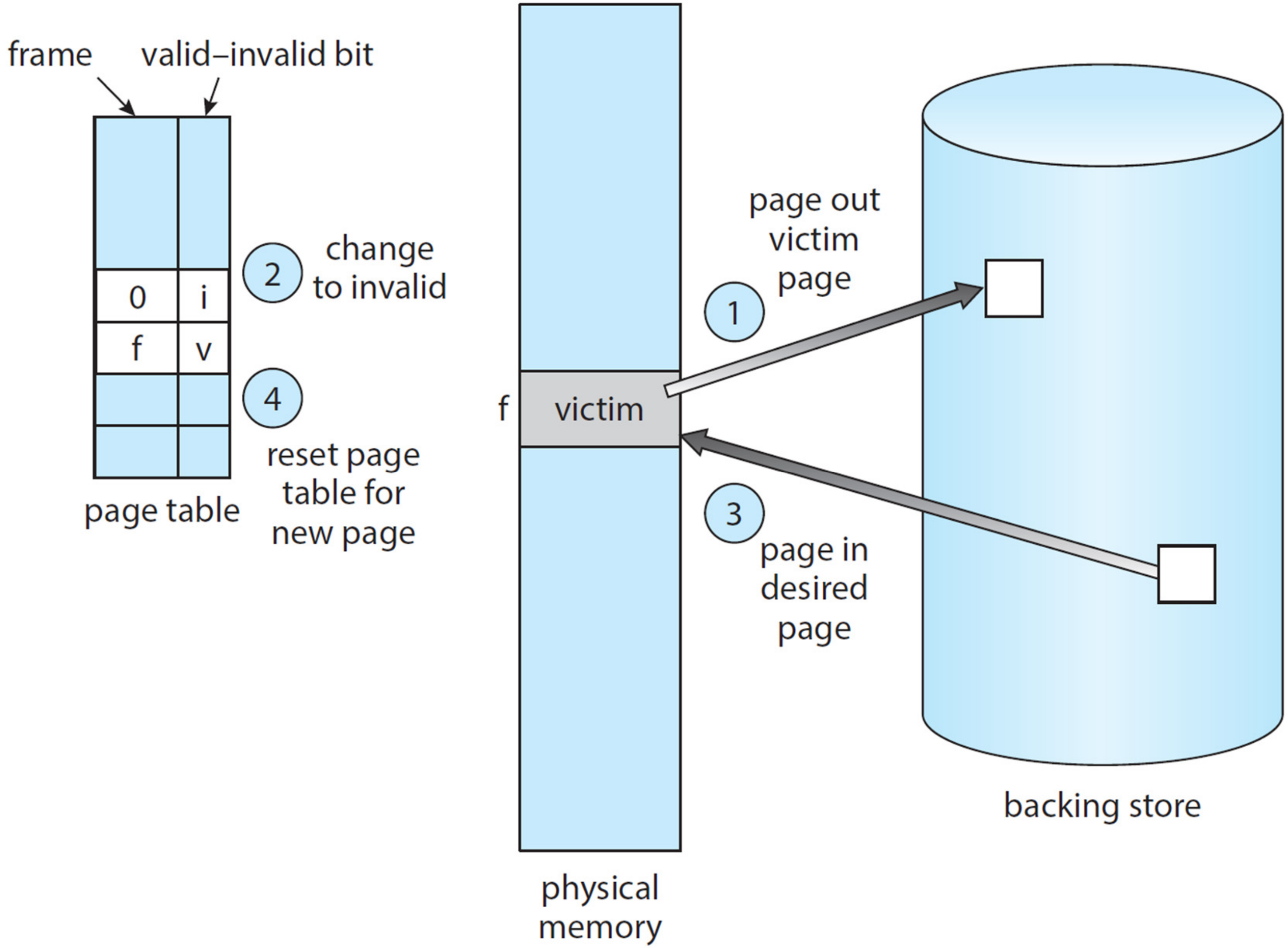
1. Find the location of the desired page on disk.

2. Find a free frame
   - If there is a free frame, use it.
   - If there is no free frame, use a page replacement algorithm to select a **victim frame.**
   - Write victim frame to disk if dirty.

3. Bring the desired page into the (newly) free frame; update the page and frame tables.

4. Continue the process by restarting the instruction that caused the trap.

Page Replacement



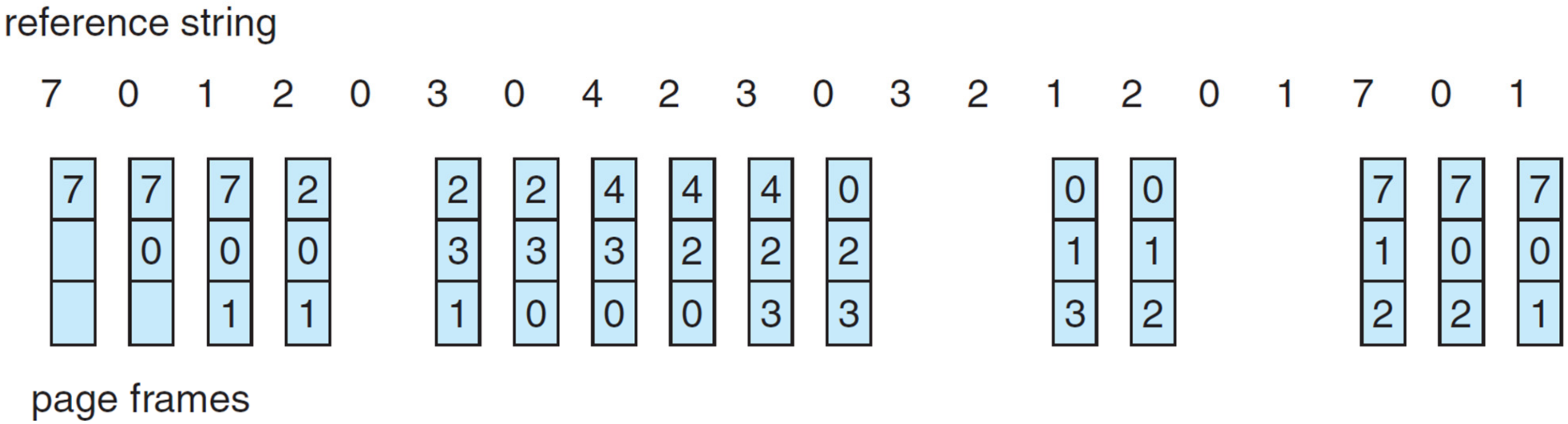
First-In-First-Out (FIFO) Algorithm

Use a FIFO queue. When a page is brought into memory, append it into the queue.

When a page must be replaced, choose the oldest page in memory (the first one in the queue).

Example: Reference string: **7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1**

3 frames (3 pages can be in memory at a time per process)

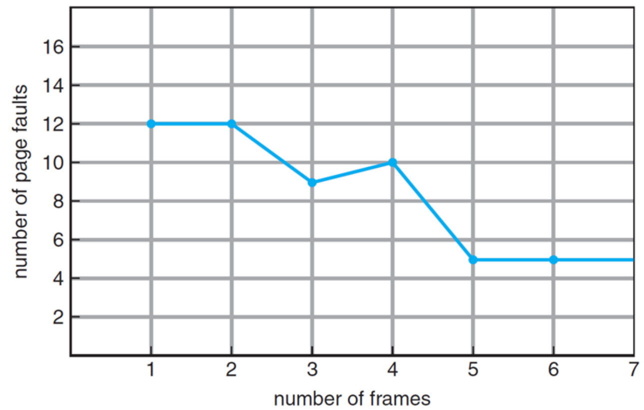


15 page faults

Adding more frames can cause more page faults!

Belady's Anomaly

consider reference string: 1,2,3,4,1,2,5,1,2,3,4,5



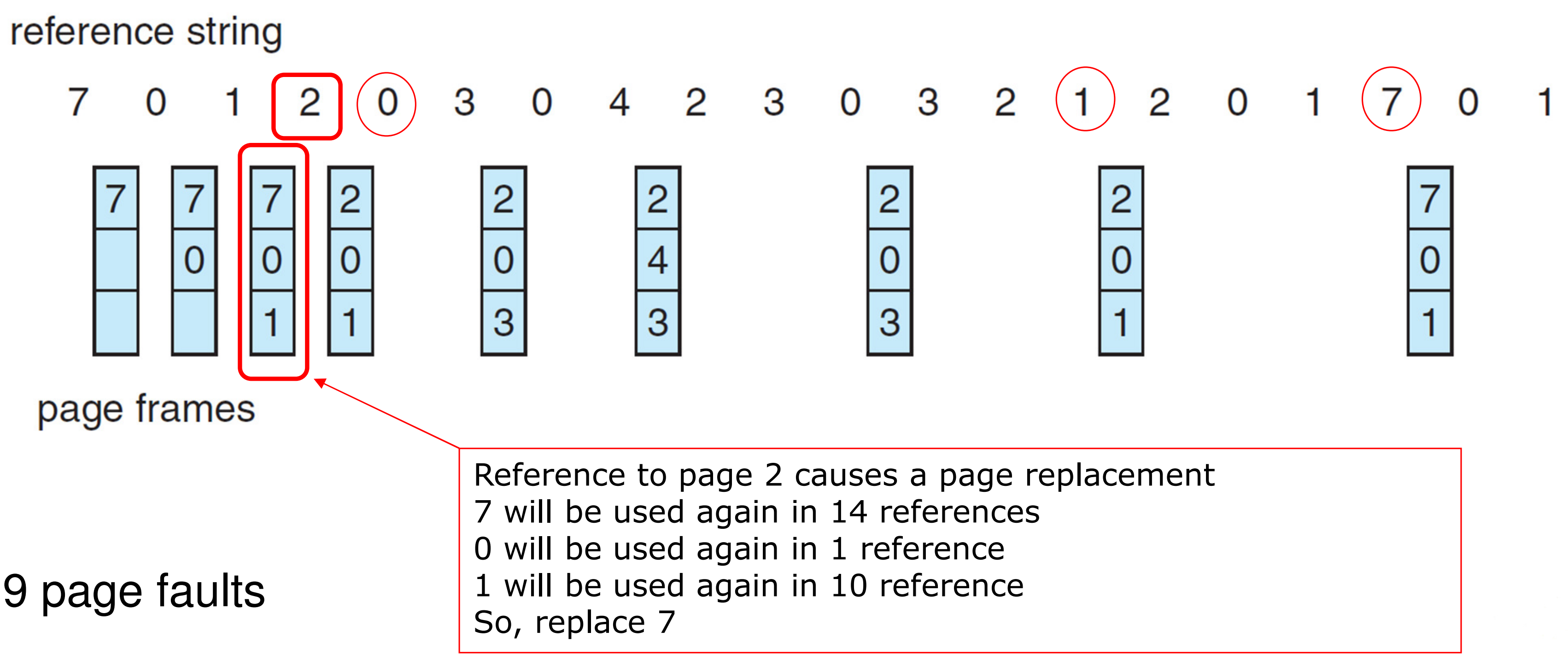
Optimal Algorithm

Replace page that will not be used for longest period of time

How do you know this?

Can’t read the future

Used for measuring how well your algorithm performs



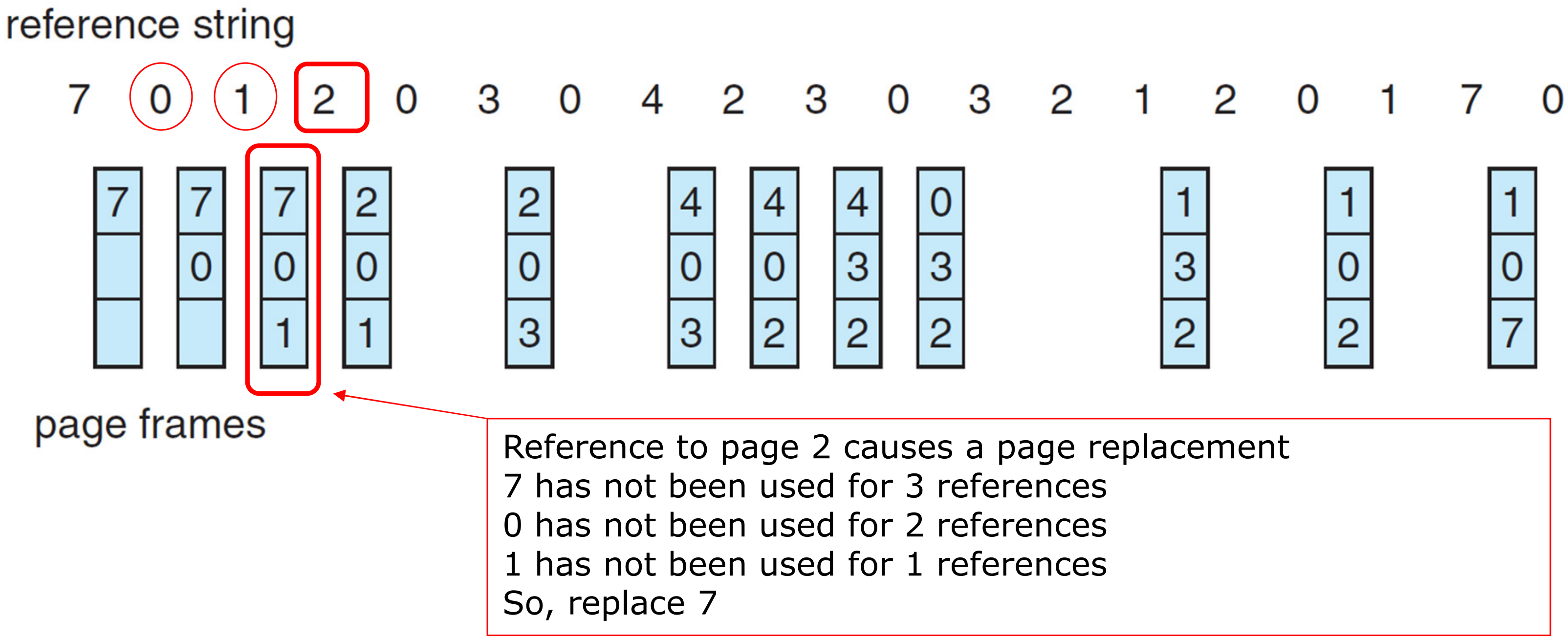
9 page faults

Least Recently Used (LRU) Algorithm

Use past knowledge rather than future

Replace page that has not been used in the most amount of time

Associate time of last use with each page



12 faults – better than FIFO but worse than Optimal Algorithm

Generally good algorithm and frequently used

But how to implement?

LRU Algorithm (Cont.)

Counter implementation

- Every page entry has a counter; every time page is referenced through this entry, copy the clock into the counter
- When a page needs to be changed, look at the counters to find smallest value
  - Search through table needed

Stack implementation

- Keep a stack of page numbers in a double link form:
- Page referenced:
  - move it to the top
  - requires 6 pointers to be changed
- But each update more expensive
- No search for replacement

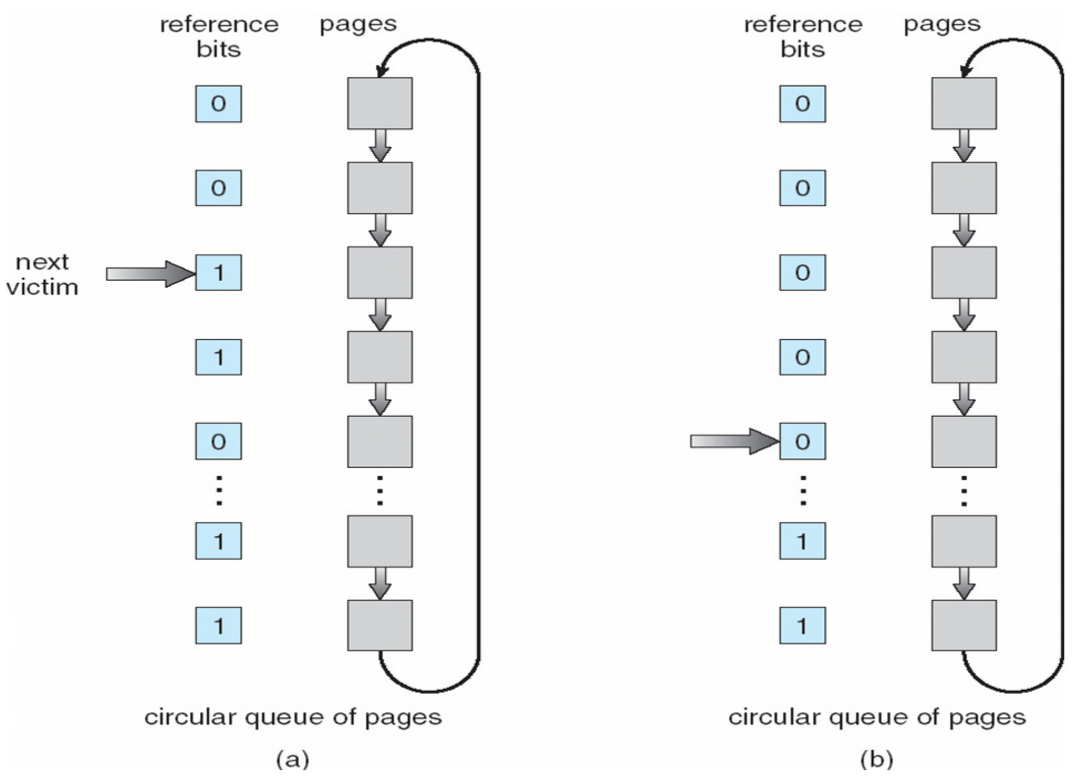
LRU and OPT are cases of **stack algorithms** that don't have Belady's Anomaly

- In a stack algorithm, the set of pages in memory for *n* frames is always a subset of the set of pages that would be in memory with *n+1* frames.

LRU Approximation Algorithms

- LRU needs special hardware and still slow
- **Reference bit**
  - With each page associate a bit, initially = 0
  - When page is referenced bit set to 1
  - Replace any with reference bit = 0 (if one exists)
    - We do not know the order, however
- **Second-chance algorithm**
  - Generally FIFO, plus hardware-provided reference bit
  - Clock replacement
  - If page to be replaced has
    - Reference bit = 0 -> replace it
    - reference bit = 1 then:
      - set reference bit 0, leave page in memory
      - replace next page, subject to same rules

Second-Chance (clock) Page-Replacement Algorithm



- Use a circular queue to keep pages in FIFO manner.
  - Oldest page in memory is at the head of queue and will be replaced first.
  - But if it has been used, it will be given a second chance.
  - Each page has a reference bit. Initialized to zero and set to one when accessed. Also reset to zero periodically.
- If page to be replaced has
  - Reference bit = 0 → replace it
  - reference bit = 1 then:
    - set reference bit 0, leave page in memory
    - replace next page, subject to same rules

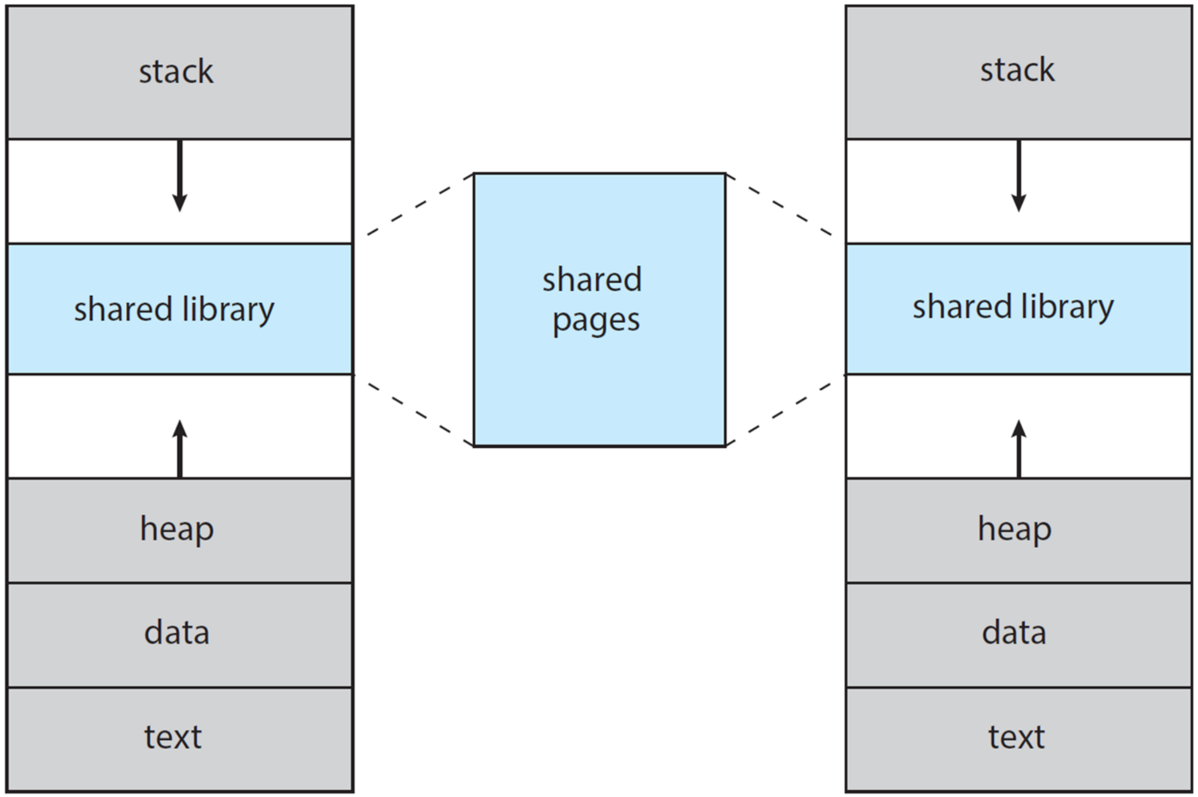
Note: Second-Chance algorithm can behave like FIFO if all pages are frequently accessed so they all get the second chance and finally the first page in the queue is the victim. Normally, this is unlikely.

Shared Pages

- Virtual memory allows pages to be shared among processes
- Logical memory pages of multiple processes are mapped to the same physical memory frames.
- This technique enables many capabilities:
    - Shared libraries
    - Memory-Mapped files
    - Shared-memory communication

Shared Library Using Virtual Memory

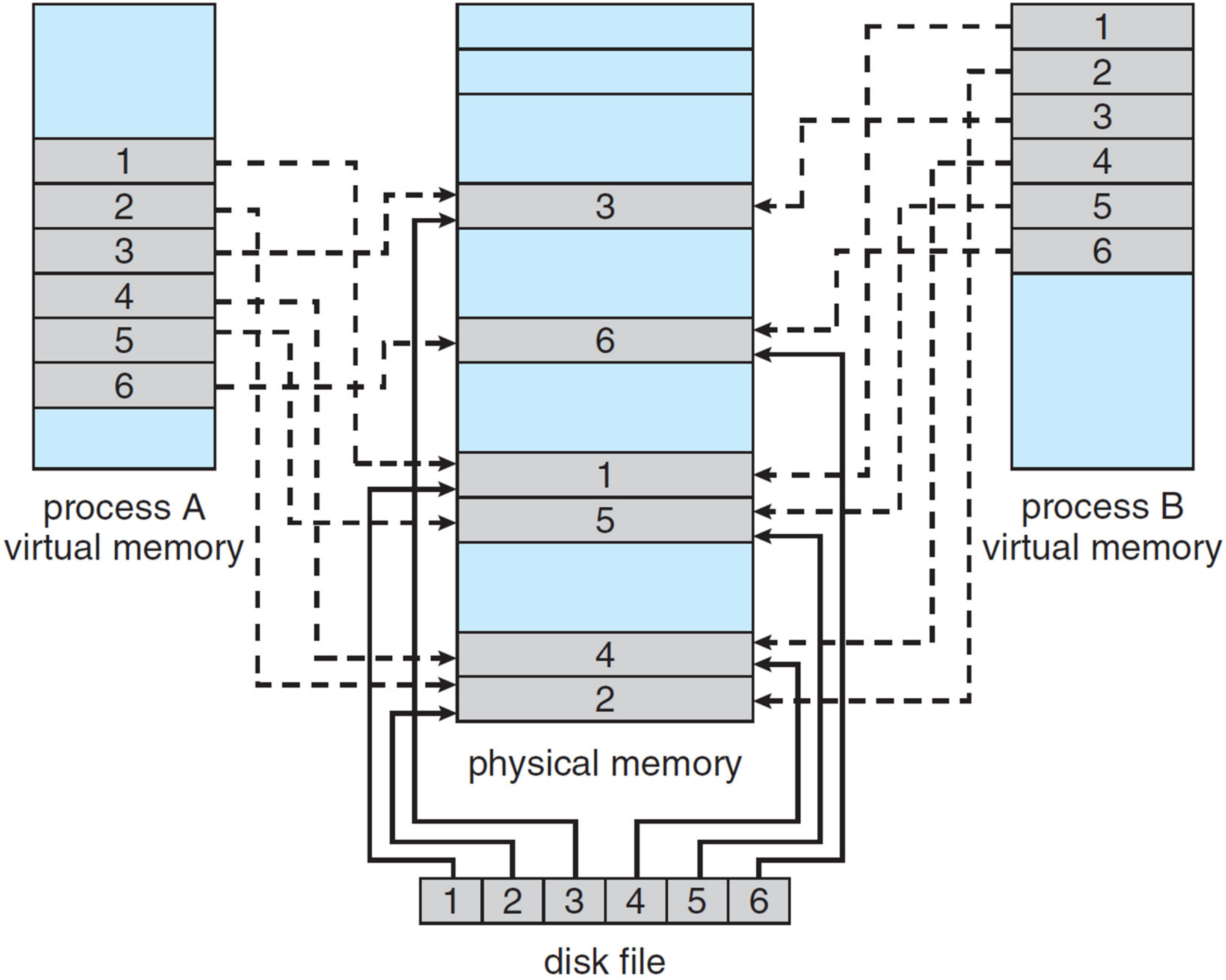
- Code libraries can be shared by several processes.
- Each process considers the library as part of its virtual address space.
- The actual pages where the libraries reside are shared.



Memory-Mapped Files

- Memory-mapped file I/O allows file I/O to be treated as routine memory access by **mapping** a disk block to a page in memory
- A file is initially read using demand paging
  - A page-sized portion of the file is read from the file system into a physical page
  - Subsequent reads/writes to/from the file are treated as ordinary memory accesses
- Simplifies and speeds file access by driving file I/O through memory rather than `read()` and `write()` system calls
- Also allows several processes to map the same file allowing the pages in memory to be shared
- But when does written data make it to disk?
  - Periodically and / or at file `close()` time

Memory Mapped Files



Memory-Mapped Shared Memory

- Processes can communicate through shared memory.
- A process can create a region of memory that it share with another process.
- Processes sharing this region consider it part of their virtual address space, yet the actual physical pages of memory are shared.